

Table 3 | Practical steps and examples to implement FUTURE-AI recommendations during design phase

Recommendations	Operations	Examples
Engage interdisciplinary stakeholders (general 1)	Identify all relevant stakeholders	Patients, GPs, nurses, ethicists, data managers
82 83	Provide information on the AI tool and AI Educational seminars, training materials, webinars	
84	Set up communication channels with stakeholders	Regular group meetings, one-to-one interviews, virtual platform
85	Organise cocreation consensus meetings	One day cocreation workshop with n=15 multidisciplinary stakeholders
86	Use qualitative methods to gather feedback	Online surveys, focus groups, narrative interviews
87	Define intended use and user requirements (usability 1)	Define the clinical need and AI tool's goal Risk prediction, disease detection, image quantification
	Define the AI tool's end users	Patients, cardiologists, radiologists, nurses
	Define the AI model's inputs	Symptoms, heart rate, blood pressure, ECG, image scan, genetic test
	Define the AI tool's functionalities and interfaces	Data upload, AI prediction, AI explainability, uncertainty estimation
88	Define requirements for human oversight	Visual quality control, manual corrections
89 90	Adjust user requirements for all end user subgroups	According to role, age group, digital literacy level
91	Define intended clinical settings and cross setting variations (universality 1)	Define the AI tool's healthcare setting(s) Primary care, hospital, remote care facility, home care
	Define the resources needed at each setting	Personnel (experience, digital literacy), medical equipment (eg, >1.5 T MRI scanner), IT infrastructure
	Specify if the AI tool is intended for high end and/or low resource settings	Facilities with MRI scanners >1.5 T v low field MRIs (eg, 0.5 T), high end v low cost portable ultrasound
92 93	Identify all cross settings variations	Data formats, medical equipment, data protocols, IT infrastructure
94	Define sources of data heterogeneity (robustness 1)	Engage relevant stakeholders to assess data heterogeneity Clinicians, technicians, data managers, IT managers, radiologists, device vendors
	Identify equipment related data variations	Differences in medical devices, manufacturers, calibrations, machine ranges (from low cost to high end)
95	Identify protocol related data variations	Differences in image sequences, data acquisition protocols, data annotation methods, sampling rates, preprocessing standards
	Identify operator related data variations	Different in experience and proficiency, operator fatigue, subjective judgment, technique variability
	Identify sources of artefacts and noises	Image noise, motion artefacts, signal dropout, sensor malfunction
	Identify context specific data variations	Lower data quality acquisition in emergency units, during high patient volume times
	Define any potential sources of bias (fairness 1)	Engage relevant stakeholders to define the sources of bias Patients, clinicians, epidemiologists, ethicists, social carers
97 98	Define standard attributes that might affect the AI tool's fairness	Sex, age, socioeconomic status
99	Identify application specific sources of bias beyond standard attributes	Skin colour for skin cancer detection, breast density for breast cancer detection
34	Identify all possible human biases	Data labelling, data curation
99	Define the need and requirements for explainability with end users (explainability 1)	Engage end users to define explainability requirements Clinicians, technicians, patients
102	Specify if explainability is necessary	Not necessary for AI enabled image segmentation part, critical for AI enabled diagnosis
	Specify the objectives of AI explainability (if it is needed)	Understanding AI model, aiding diagnostic reasoning, justifying treatment recommendations
103	Define suitable explainability approaches	Visual explanations, feature importance, counterfactuals
104	Adjust the design of the AI explanations for all end user subgroup	Heatmaps for clinicians, feature importance for patients
105 106	Investigate ethical issues (general 6)	Consult ethicists on ethical considerations
	Ethicists specialised in medical AI and/or in the application domain (eg, paediatrics)	
107	Assess if the AI tool's design is aligned with relevant ethical values	Right to autonomy, information, consent, confidentiality, equity
107	Identify application specific ethical issues	Ethical risks for a

paediatric AI tool (eg, emotional impact on children) 108 109 Comply with local ethical AI frameworks AI ethical guidelines from Europe,2 United Kingdom,110 111 United States,112 Canada,113 China,114 India,115 Japan,116 117 Australia,118 etc Investigate social and environmental issues (general 7) Investigate AI tool's social and environmental impact Workforce displacement, worsened working conditions and relations, deskilling,80 dehumanisation of care, reduced health literacy, increased carbon footprint,119 negative public perception107 120 Define mitigations to enhance the AI tool's social and environmental impact Interfaces for physician-patient communication, workforce training, educational programmes, energy efficient computing practices, public engagement initiatives Optimise algorithms, energy efficiency Develop and use energy efficient algorithms that minimise computational demands. Techniques like model pruning, quantisation, and edge computing can reduce the energy required for AI tasks Promote responsible data usage Focus on collecting and processing only the necessary amount of data. Implement federated learning techniques to minimise data transfers. This approach keeps data localised, reducing need for extensive data movement, which consumes energy Monitor and report the environmental impact of the AI tool Regularly monitor and report on the environmental impact of AI systems used in healthcare, including energy usage, carbon emissions, and waste generation Use community defined standards (universality 2) Use a standard definition for the clinical task Definition of heart failure by the American Academy of Cardiology 121 Use a standard method for data labelling BI-RADS for breast imaging122 Use a standard ontology for the AI inputs DICOM for imaging data,123 SNOMED for clinical data41 Adopt technical standards IEEE 2801-2022 for medical AI software43 Use standard evaluation criteria See Maier-Hein et al21 for medical imaging applications, Barocas et al31 and Bellamy et al35 for fairness evaluation Implement a risk management process (traceability 1) Identify all possible clinical, technical, ethical, and societal risks Bias against under-represented subgroups, limited generalisability to low resource facilities, data drift, lack of acceptance by end users, sensitivity to noisy inputs 124 Identify all possible operational risks Misuse of the AI tool (owing to insufficient training or not following the instructions), application of the AI tool outside of the target population (eg, individuals with implants), use of the tool by others than the target end users (eg, technician instead of physician), hardware failure, incorrect data annotations, adversarial attacks 76 125 Assess the likelihood of each risk Very likely, likely, possible, rare Assess the consequences of each risk Patient harm, discrimination, lack of transparency, loss of autonomy, patient reidentification126 Prioritise all the risks depending on their likelihood and consequences Risk of bias (if no personal attributes are included in the model) v risk of patient reidentification (if personal attributes are collected) Define mitigation measures to be applied during AI development Data enhancement, data augmentation,127 bias correction techniques, domain adaptation,66 transfer learning,128 continuous learning129 Define mitigation measures to be applied after deployment Warnings to the users, system shutdown, reprocessing of the input data, acquisition of new input data, use of an alternative procedure, or human judgment only Set up a mechanism to monitor and manage risks over time Periodic risk assessment every six months Create a comprehensive risk management file Including all risks, their likelihood and consequences, risk mitigation measures, risk monitoring strategy

Collect representative training dataset (robustness 2) Collect training data that reflect the demographic variations According to age, sex, ethnicity, socioeconomics Collect training data that reflect the clinical variations Disease subgroups, treatment protocols, clinical outcomes, rare cases Collect training data that reflect variations in real world practice Data acquisition protocols, data annotations, medical equipment, operational variations (eg, patient motion

during scanning) 125 Artificially enhance the training data to mimic real world conditions Data augmentation,127 data synthesis (eg, low quality data, noise addition),130 data harmonisation,131 132 data homogenisation133 Collect information on individuals' and data attributes (fairness 2) Request approval for collecting data on personal attributes Sex, age, ethnicity, socioeconomic status 134 Collect information on standard attributes of the individuals (if available and allowed) Sex, age, nationality, education135 Include application specific information relevant for fairness analysis Skin colour, breast density,34 presence of implants, comorbidity 136 Estimate data distributions across subgroups Male v female, across ethnic groups Collect information on data provenance Data centres, equipment characteristics, data preprocessing, annotation processes Implement measures for data privacy and security (general 2) Implement measures to ensure data privacy and security Data deidentification, federated learning,137 138 differential privacy, encryption139 Implement measures against malicious attacks Firewalls, intrusion detection systems, regular security audits 139 Adhere to applicable data protection regulations General Data Protection Regulation,140 Health Insurance Portability and Accountability Act 141 Define suitable data governance mechanisms Access control, logging system Implement measures against identified AI risks (general 3) Implement a baseline AI model and identify its limitations Bias, lack of generalisability 142 Implement methods to enhance robustness to real world variations (if needed) Regularisation,143 data augmentation,127 data harmonisation,131 domain adaptation66 Implement methods to enhance generalisability across settings (if needed) Regularisation, transfer learning,144 knowledge distillation145 Implement methods to enhance fairness across subgroups (if needed) Data resampling, bias free representation,36 equalised odds postprocessing37 38 146 Establish mechanisms for human-AI interactions (usability 2) Implement mechanisms to standardise data preprocessing and labelling Data preprocessing pipeline, data labelling plugin Implement an interface for using the AI model Application programming interface Implement interfaces for explainable AI Visual explanations, heatmaps, feature importance bars 105 106 Implement mechanisms for user centred quality control of the AI results Visual quality control, uncertainty estimation147 Implement mechanism for user feedback Feedback interface148

Recommendations Operations Examples Define adequate evaluation plan (general 4) Identify the dimensions of trustworthy AI to be evaluated Robustness, clinical safety, fairness, data drifts, usability, explainability Select appropriate testing datasets External dataset from a new hospital, public benchmarking dataset 148 Compare the AI tool against standard of care Conventional risk predictors, visual assessment by radiologist, decision by clinician149 150 Select adequate evaluation metrics F1 score for classification, concordance index for survival,21 statistical parity for fairness 151 Evaluate using external datasets and/or multiple sites (universality 3) Identify relevant public datasets Cancer Imaging Archive,152 UK Biobank,153 M&Ms,154 MAMA-MIA,155 BRATS156 Identify external private datasets New prospective dataset from same site or from different clinical centre157 158 Select multiple evaluation sites Three sites in same country, five sites in two different countries Verify that evaluation data and sites reflect real world variations Variations in demographics, clinicians, equipment Confirm that no evaluation data were used during training Yes/no Evaluate fairness and bias correction measures (fairness 3) Select attributes and factors for fairness evaluation Sex, age, skin colour, comorbidity Define fairness metrics and criteria Statistical parity difference defined fairness between -0.1 and 0.1 135 Evaluate fairness and identify biases Fair with respect to age, biased with respect to sex Evaluate bias mitigation measures Training data resampling,159 equalised odds postprocessing37 38 146 Evaluate impact of mitigation measures on model performance Data resampling removed sex bias but reduced model

performance¹⁶⁰ Report identified and uncorrected biases In AI information leaflet and technical documentation¹⁶¹ (see traceability 2). Evaluate user experience (usability 4) Evaluate usability with diverse end users According to sex, age, digital proficiency level, role, clinical profile¹⁶² ¹⁶³ Evaluate user satisfaction using usability questionnaires System usability scale¹⁶⁴ Evaluate user performance and productivity Diagnosis time with and without AI tool, image quantification time¹⁶⁵ Assess training of new end users Average time to reach competency, training difficulties ¹⁶⁶ Evaluate clinical utility and safety (usability 5) Define clinical evaluation plan Randomised control trial,⁵⁹ ¹⁶⁷ in silico trial ¹⁶⁸ Evaluate if AI tool improves patient outcomes Better risk prevention, earlier diagnosis, more personalised treatment ¹⁶⁹ Evaluate if AI tool enhances productivity or quality of care Enhanced patient triage, shorter waiting times, faster diagnosis, higher patient intake¹⁶⁹ Evaluate if AI tool results in cost savings Reduction in diagnosis costs,¹⁷⁰ ¹⁷¹ reduction in overtreatment ¹⁷² Evaluate AI tool's safety Side effects or major adverse events in randomised control trials ¹⁷³ ¹⁷⁴ Evaluate robustness (robustness 3) Evaluate robustness under real world variations Using test-retest datasets,¹⁷⁵ ¹⁷⁶ multivendor datasets ¹⁷⁷ Evaluate robustness under simulated variations Using simulated repeatability tests,¹⁴⁸ synthetic noise and artefacts (eg, image blurring) ¹⁷⁸ Evaluate robustness against variations in end users Different technicians or annotators Evaluate mitigation measures for robustness enhancement Regularisation,⁶³ data augmentation,⁶⁴ ¹²⁷ noise addition, normalisation,¹⁷⁹ resampling, domain adaptation⁶⁶ Evaluate explainability (explainability 2) Assess if explanations are clinically meaningful Reviewing by expert panels, alignment to current clinical guidelines, explanations not pointing to shortcuts ⁷³ Assess explainability quantitatively using objective measures Fidelity, consistency, completeness, sensitivity to noise¹⁸⁰⁻¹⁸² Assess explainability qualitatively with end users Using user tests or questionnaires to measure confidence and affect clinical decision making¹⁸³ ¹⁸⁴ Evaluate if explanations cause end user overconfidence or overreliance Measure changes in clinician confidence,¹⁸⁵ ¹⁸⁶ performance with and without AI tool ¹⁸⁷ Evaluate if explanations are sensitive to input data variations Stress tests under perturbations to evaluate the stability of explanations ⁷⁴ ¹⁸⁸ Provide documentation (traceability 2) Report evaluation results in publication using AI reporting guidelines Peer reviewed scientific publication using TRIPOD-AI reporting guideline¹⁵ Create technical documentation for AI tool AI passport,¹⁸⁹ model cards⁴⁹ (including model hyperparameters, training and testing data, evaluations, limitations, etc) Create clinical documentation for AI tool Guidelines for clinical use, AI information leaflet (including intended use, conditions and diseases, targeted populations, instructions, potential benefits, contraindications) Provide risk management file Including identified risks, mitigation measures, monitoring measures Create user and training documentation User manuals, training materials, troubleshooting, FAQs (see usability 2) Identify and provide all locally required documentation Compliance documents and certifications (see general 5)

Evaluate and demonstrate local clinical validity (universality 4) Test AI model using local data Data from local clinical registry Identify factors that could affect AI tool's local validity Local operators, equipment, clinical workflows, acquisition protocols Assess AI tool's integration within local clinical workflows AI tool's interface aligns with hospital IT system¹⁴⁸ or disrupts routine practice Assess AI tool's local practical utility and identify any operational challenges Time to operate, clinician satisfaction, disruption of existing operations ¹⁴⁸ ¹⁹⁰ Implement adjustments for local validity Model calibration, fine-tuning,¹⁹¹ transfer learning¹⁹²⁻¹⁹⁴ Compare performance of AI tool with that of local clinicians Side-by-side comparison, in silico trial Define mechanisms for quality control of AI inputs and outputs (traceability 3) Implement mechanisms to identify erroneous input data Missing value or out-of-distribution detector,¹⁹⁵ automated image quality assessment ⁷³ ¹⁹⁶ ¹⁹⁷ Implement mechanisms to detect implausible

AI outputs Postprocessing sanity checks, anomaly detection algorithm¹⁹⁸ Provide calibrated uncertainty estimates to inform on AI tool's confidence Calibrated uncertainty estimates per patient or data point^{53 54 199} Implement system for continuous quality monitoring Real time dashboard tracking data quality and performance metrics²⁰⁰ Implement feedback mechanism for users to report issues Feedback portal enabling clinicians to report discrepancies or anomalies Implement system for periodic auditing and updating (traceability 4) Define schedule for periodic audits Biannual or annual Define audit criteria and metrics Accuracy, consistency, fairness, data security¹⁴⁸ Define datasets for periodic audits Newly acquired prospective dataset from local hospital Implement mechanisms to detect data or concept drifts Detecting shifts in input data distributions^{148 190} Assign role of auditor(s) for AI tool Internal auditing team, third party company¹⁹⁰ Update AI tool based on audit results Updating AI model,⁵⁶ re-evaluating AI model,¹⁴⁸ adjusting operational protocols, continuous learning²⁰¹⁻²⁰⁴ Implement reporting system from audits and subsequent updates Automatic sharing of detailed reports to healthcare managers and clinicians Monitor impact of AI updates Impact on system performance and user satisfaction⁵⁶ Implement logging system for usage recording (traceability 5) Implement logging framework capturing all interactions User actions, AI inputs, AI outputs, clinical decisions Define data to be logged Timestamp, user ID, patient ID (anonymised), action details, results Implement mechanisms for data capture Software to automatically record every data and operation Implement mechanisms for data security Encrypted log files, privacy preserving techniques²⁰⁵ Provide access to logs for auditing and troubleshooting By defining authorised personnel, eg, healthcare or IT managers Implement mechanism for end users to log any issues A user interface to enter information about operational anomalies Implement log analysis Time series statistics and visualisations to detect unusual activities and alert administrators Provide training (usability 3) Create user manuals User instructions, capabilities, limitations, troubleshooting steps, examples, and case studies Develop training materials and activities Online courses, workshops, hands-on sessions Use formats and languages accessible to intended end users Multiple formats (text, video, audio) and languages (English, Chinese, Swahili) Customise training to all end user groups Role specific modules for specialists, nurses, and patients Include training to enhance AI and health literacy On application specific AI concepts (eg, radiomics, explainability), AI driven clinical decision making Identify and comply with applicable AI regulatory requirements (general 5) Engage regulatory experts to investigate regulatory requirements Regulatory consultants from intended local settings Identify specific regulations based on AI tool's intended markets FDA's SaMD in the United States,²⁰⁶ MDR and AI Act²⁰⁷ in the EU Identify specific requirements based on AI tool's purpose De Novo classification (Class III)²⁰⁸ Define list of milestones towards regulatory compliance MDR certification: technical verification, pivotal clinical trial, risk and quality management, postmarket follow-up Establish mechanisms for AI governance (traceability 6) Assign roles for AI tool's governance For periodic auditing, maintenance, supervision (eg, healthcare manager) Define responsibilities for AI related errors Responsibilities of clinicians, healthcare centres, AI developers, and manufacturers Define mechanisms for accountability Individual v collective accountability/liability,²⁵ compensations, support for patients